

Yellow Bay

Process Integration Lead · Ardley Fab 7

15 DAYS · 49 STOPS · GRADE 12 · YELLOWBAY

THE OPENING CARD

Since March, four wafers in ten come out of Ardley Fab 7 too poor to sell. Nobody agrees why. You are the process integration lead, and the proving batch file that goes with the next lot is yours to sign. In fifteen days that lot runs, and one customer buys sixty per cent of all Ardley makes. That customer reads the file, then keeps the contract or drops a plant with nine hundred people in it.

What the wafer is, before anybody argues about it

BEFORE THE DAY — GREET

Twelve people, two wings, one floor

Twelve people work these two wings, so get round as many of them as you can now, while nobody is under pressure and no clock is running on any of it.

PLAN CARD 157W

Monday. The qualification lot runs in fifteen days, whatever anybody has worked out by then. The file that goes out with it has to say three things. What is in the chamber. What the film is made of. Which step decides the yield. Simone Hale, the fab manager, has stopped taking explanations that are not measurements. She wants numbers. Hana Ferreira, who runs surface analysis, wants every incoming lot held at the door. Ruben Ostrowski, who runs deposition, wants the line running while the recipes are checked. Both cannot happen. Today you count how many atoms are really in one lot of wafers. You also settle what makes one atom a different element from another. Then you read a purity limit for what it claims. Toby Whelan runs the furnace on nights, and his son Danny is four months into the gown room. None of the arguing has produced a number either of them can act on.

BRIEFING 12W

Four wafers in ten are failing and two leads have opposite explanations.

WHAT YOU DECIDE 14W

Establish what is actually on the floor and what the paperwork claims about it.

1.1 What is actually in a lot

WAFER STORE • A ROOM • BALLPARK

SCENE 37W

Marcus Twill, the incoming materials lead, has a lot of twenty-five wafers on the bench and a certificate that quotes everything as a fraction. Hale wants the impurity argument in atoms, and nobody has converted anything yet.

GUIDE 61W

The tiles hold everything on the bench and two quantities that belong to a different question. A mass on a balance and a molar mass are what turn grams into a count. The thickness and the number of wafers in the box describe the lot rather than one wafer. Pick the two that answer what was asked about a single wafer.

ASK 9W

How many moles of silicon are in one wafer?

TAKES AS READ

a balance reads mass and a reaction happens in atoms • silicon is a single element with one molar mass

VERDICT 90W

A balance reads grams and everything that matters here happens atom by atom, so the division by molar mass is the step that makes the two commensurable. Four and a half moles is 2.7×10^{24} silicon atoms in one disc, which is the scale any impurity figure has to be read against. The thickness and the lot count are real numbers about real things, and neither of them enters this. They describe the geometry of the wafer and the size of the delivery, not the amount of substance weighed.

TAKEAWAY 24W

A mass divided by a molar mass is a count of particles, and it is the only route from a balance to an atom.

1.2 The one number that names an element

METROLOGY BAY · A PERSON · CHOICE

SCENE 42W

Ferreira has two spectra on the screen from two wafers that weigh the same, look the same and came out of the same box. Something in one of them is not silicon and the argument has already started about what that means.

GUIDE 46W

Four differences between two atoms are on offer and only one of them changes what the atom is called. The test is which quantity you could change while leaving the element the same. Getting this wrong is how a mass measurement gets read as an identification.

ASK 9W

Which difference between two atoms makes them different elements?

TAKES AS READ

an atom is a nucleus of protons and neutrons with electrons around it

VERDICT 86W

The nuclear charge is what an element is. Change the neutrons and you have another isotope of the same element, with the same chemistry and a different mass. Change the electrons and you have an ion, which is the same element carrying a charge. Rearrange the outer electrons and you have an excited or bonded state of it. Only the proton count is unable to change without the substance becoming something else, which is why the periodic table is ordered by it and not by mass.

TAKEAWAY 17W

The proton count fixes the element; everything else about an atom can vary without changing its name.

1.3 What five parts in a billion is a claim about

IMPLANT BAY · A ROOM · CHOICE

SCENE 37W

The incoming specification for substrate silicon sets a limit of five parts per billion on boron. Priyanka Ganesan, the implant process lead, has the certificates for six lots on the desk and every one of them passes.

GUIDE 62W

The four readings of that limit differ in what they claim about. One is a ratio, one a mass, one a count fixed once and for all, one a grade against other suppliers. Only one survives being applied to a piece of any size, and that is the test. The distinction decides whether a passing certificate is evidence about a particular wafer.

ASK 10W

What does a limit of five parts per billion constrain?

TAKES AS READ

a fraction is a comparison between two amounts of the same kind

VERDICT 93W

A part per billion is a comparison, not a quantity. It says that for every thousand million atoms, no more than five are the impurity. That stays true of a wafer, a lot or a shaving off the edge. That is what makes it useful as a specification, and what makes it deceptive as a picture. A wafer holding four and a half moles of silicon meets that limit while carrying about thirteen million billion boron atoms. The ratio is small and the count is not, and the lattice responds to the count.

TAKEAWAY 22W

A parts-per-billion figure is a ratio, so the number of impurity atoms it permits depends on how much material you are holding.

END OF THE DAY 19W

Every claim in this building reduces to a count of atoms, and a count is something you can check.

Light with a threshold, and gas that never stops moving

PLAN CARD 132W

Tuesday. The first re-run of the failing layer goes into the yellow bay this afternoon. Yuki Tanabe, who runs lithography, has been asked twice this week why her bay is lit amber. Twice she has given an answer that satisfied nobody. That matters more than it sounds. The same argument decides which lamps she may change. Ostrowski wants the deposition chamber opened between runs and closed again. That would save four hours a run. Today you settle what light of one wavelength can and cannot do to a wafer. You work out what the gas inside a chamber is doing while nothing seems to happen. Then you look at how tightly an atom holds the electrons that decide both. Four hours a run adds up to a whole day of the fifteen.

BRIEFING 15W

The yellow bay is lit amber for a reason nobody on the shift can state.

WHAT YOU DECIDE 18W

Establish what the light in this building can do and what the gas in the chambers is doing.

2.1 Why the bay is lit amber

YELLOW BAY · A ROOM · BALLPARK

SCENE 42W

Tanabe has the exposure lamp at 248 nanometres and the room lit at 580. Park, the track operator, wants to know why a brighter safelight would not do just as much harm as a dim exposure, and nobody has done the arithmetic.

GUIDE 51W

Two wavelengths are on the tiles and only one of them belongs in the exposure calculation. The constants are fixed and the wavelength is the variable, so the whole question is which length you put underneath the line. Brightness is not on the tiles at all, which is itself the finding.

ASK 11W

How much energy does one photon of the exposure light carry?

TAKES AS READ

light of a given wavelength arrives in packets of a fixed energy · a nanometre is a thousand-millionth of a metre

VERDICT 91W

Energy per photon is set by wavelength alone, and the resist responds to single photons rather than to the total energy landing on it. The exposure photon carries about eight times ten to the minus nineteen joules. The amber one, at more than twice the wavelength, carries less than half of that. So it falls under the threshold the resist chemistry needs. That is why the brightness of the safelight is close to irrelevant and its colour is not. More photons that individually cannot do the job still cannot do it.

TAKEAWAY 28W

The energy of a photon rises as its wavelength falls, so the wavelength decides what light can do and the brightness only decides how much of it arrives.

2.2 What a gas is doing while nothing is happening

DEPOSITION BAY · A PERSON · SEQUENCE

SCENE 39W

Lena Vance, the gas and chemicals supervisor, has four cylinders feeding the chamber and a leak that shows up as one gas arriving somewhere it should not. Ostrowski wants to know which of the four would get there first.

GUIDE 48W

At one temperature every molecule in the chamber has the same average energy, and energy is shared between mass and speed. A heavy molecule therefore carries that energy as mass and a light one as speed. Rank them by the mass on the label and the order follows.

ASK 10W

Order the four gases in the chamber by average speed.

TAKES AS READ

temperature is a measure of the average kinetic energy of the particles · a molar mass can be read off a chemical formula

VERDICT 91W

Temperature fixes the average kinetic energy, which is a half of the mass times the speed squared, so at one temperature a heavier molecule must be slower. Argon is twenty times the mass of hydrogen and moves about four and a half times more slowly. The speed goes as the square root of the mass rather than as the mass. That square root is why the spread in speeds is far narrower than the spread in masses. It is also why hydrogen reaches a leak first without being twenty times quicker.

TAKEAWAY 18W

At one temperature all molecules carry the same average energy, so the light ones must be moving fastest.

2.3 How tightly the nucleus holds on

METROLOGY BAY • A ROOM • BALLPARK

SCENE 42W

Aaron Dorsey, the metrology technician, has a table on the bench. It gives the energy needed to knock an electron out of each element on the wafer. Ferreira wants to know why that energy is so much larger for some of them.

GUIDE 51W

The attraction between two charges depends on both charges and on the distance twice over. Two of the tiles describe the distance. One is the distance itself and one is its square, and only one of those belongs underneath the line. Choosing wrongly between them is the whole of the arithmetic.

ASK 10W

How strongly is silicon's outer electron attracted, in relative units?

TAKES AS READ

a nucleus carries one positive charge per proton • an electron carries one negative charge

VERDICT 89W

Attraction between charges is the product of the charges divided by the square of the distance. So moving an electron twice as far out weakens the hold by a factor of four rather than two. The number that comes out here is an overestimate, and deliberately so. It uses all fourteen protons, and an outer electron in a real silicon atom does not feel all fourteen. What sits between it and the nucleus is the subject of the next stop, and this arithmetic is what it has to correct.

TAKEAWAY 18W

Attraction rises with the charges and falls with the square of the separation, so distance costs twice over.

END OF THE DAY 19W

Light delivers energy in packets with a threshold, and a gas is particles whose speeds follow from their masses.

What the inner shells do to everything outside them

PLAN CARD 123W

Wednesday. The first data from the re-qualification comes back at four. Yesterday's sums say something odd. They say silicon should hold its outer electron about eleven hundred times more tightly than hydrogen holds its own. The measured energy is nothing like that. Dorsey has the table of measured values. Ferreira has the sums. Neither of them will move until the gap is explained. Today you work out what sits between an outer electron and the nucleus. You read a table of energies, one electron pulled off at a time, for what it says about shells. Then you settle how deep a probe may go and still be reading the surface. The argument over that gap does not end until all three are settled.

BRIEFING 13W

The attraction calculated yesterday is too large for every element on the wafer.

WHAT YOU DECIDE 11W

Correct yesterday's arithmetic for the electrons that sit in the way.

3.1 The charge that is left over

METROLOGY BAY • A ROOM • BALLPARK

SCENE 34W

Ferreira has a phosphorus atom on the board with its fifteen protons written up and ten of its electrons circled. Dorsey's measured ionisation energy is a fifth of what the full nuclear charge predicts.

GUIDE 57W

An outer electron is pulled by the nucleus and pushed by every electron between it and the nucleus, and the two nearly cancel. What is left over is a much smaller charge than the proton count. The tiles hold the protons, the inner electrons and the outer ones, and only two of the three enter the subtraction.

ASK 9W

What effective nuclear charge does phosphorus's outer electron feel?

TAKES AS READ

electrons in filled inner shells lie between the nucleus and the outer electrons • like charges repel

VERDICT 92W

Ten electrons in the filled inner shells sit between the nucleus and the outer shell. They cancel ten of the fifteen positive charges, so an outer electron responds to about five rather than fifteen. Electrons in the same shell are at roughly the same distance, and they shield each other hardly at all. That is why the outer five are left out of the subtraction. This is also why effective charge rises across a row while the shell stays the same: protons are being added and the shielding is not keeping up.

TAKEAWAY 22W

Effective nuclear charge is the proton count minus the electrons closer in, and it is what an outer electron actually responds to.

3.2 Where the ladder jumps

IMPLANT BAY • A PERSON • PROTOCOL

SCENE 35W

Ganesan has four unlabelled elements on the screen, each with the energy to remove its first six electrons in order. The supplier will not say which is which and the implant recipe depends on knowing.

GUIDE 49W

Removing electrons from one shell costs a little more each time, because the charge left behind grows. Reaching into the shell beneath costs enormously more, and where that jump falls is a count of how many electrons the outer shell held. Match each ladder to what its jump says.

ASK 13W

Match each ladder of ionisation energies to what it says about the atom.

TAKES AS READ

removing an electron from a positive ion is harder than from a neutral atom • inner shells lie closer to the nucleus than outer ones • effective nuclear charge, and shielding by the inner shells — taken as read

VERDICT 83W

Within one shell each electron leaves behind a more positive ion, so each removal costs a little more than the last. Reaching into a filled shell underneath means fighting a much larger effective charge at a much shorter distance. Both terms push the same way, so the cost multiplies rather than creeping up. Counting the gentle steps before that jump counts the outer shell directly. That is the measurement that separates a donor from an acceptor without knowing anything else about the element.

TAKEAWAY 23W

A large jump in successive ionisation energies marks the point where the next electron has to come out of a shell closer in.

3.3 The top nanometre, and everything under it

WET BENCH · A ROOM · CHOICE

SCENE 33W

Delphine Aubrey, the wet process and metallisation lead, has a bulk purity result on one wafer reading clean. A failure map on the same wafer shows edge losses. Both measurements were done properly.

GUIDE 54W

The four options differ in what part of the wafer each measurement averages over. A wafer is nearly a millimetre thick, and the layers that matter are a few atoms deep. So the ratio between the two is about a million to one. The question is what that ratio does to a contamination figure.

ASK 15W

Why can a wafer pass a bulk purity test and still fail because of contamination?

TAKES AS READ

a contamination figure is an average over whatever the instrument sampled

VERDICT 92W

Every measurement has a depth, and a bulk method reports the total divided by everything it sampled. A wafer is about three-quarters of a millimetre thick, and a device layer is a few nanometres. So a contaminant confined to the surface is being divided by roughly a million times more silicon than it actually sits on. It disappears into the average while remaining, where it is, at a concentration high enough to ruin every chip on the disc. Depth of measurement is a property of the instrument and has to be chosen.

TAKEAWAY 28W

A bulk measurement divides a surface contaminant by a million times more material than it sits on, so a clean bulk result says almost nothing about the surface.

3.4 What is actually in a lot — Review

WAFER STORE · A ROOM · CALLBACK · BALLPARK

SCENE 40W

Twill has a quartz carrier ring off one of the furnaces on the store balance and a supplier claim about how much silica a ring holds. Hale wants the claim checked as a count before the next order is signed.

GUIDE 56W

Two molar masses are on the tiles: one for a whole formula unit, one for a single element in it. They answer different questions about the same piece of glass. The other two tiles describe the delivery rather than the ring that was weighed. Pick the mass and the molar mass that belong to one another.

ASK 10W

How many moles of silica are in one carrier ring?

TAKES AS READ

a balance reads mass and a specification is written in atoms · a formula unit of silica holds one silicon atom and two oxygen atoms

VERDICT 87W

A balance reads grams. Everything a specification says happens atom by atom, so the molar mass is the step that makes the two commensurable. What was weighed is silicon dioxide. Dividing by silicon's molar mass alone counts something that was never on the pan — a larger number, about a substance the ring is not made of. Twenty-five moles of silica is about one and a half times ten to the twenty-fifth formula units. That is the scale any contamination figure for this ring is read against.

TAKEAWAY 23W

Divide by the molar mass of what was actually put on the balance, not of the element you happen to be interested in.

END OF THE DAY 22W

An outer electron is held by what is left of the nuclear charge once the inner electrons have cancelled part of it.

What is in the chamber, and what a peak at 28 could be

BEFORE THE DAY — FOLLOW

Aubrey walks the wet bench line once

Delphine Aubrey, who runs the wet benches and the metal fill, walks her line once a week and does not slow down for anybody. The sequence from strip to rinse to dry is the part of this process that no diagram records at the level that matters. Stay with her and keep clear of the tanks.

PLAN CARD 113W

Thursday. Vance has the chamber down for eight hours. It is the only window this week to find out what is inside it. The residual gas analyser shows a peak at mass twenty-eight. Ostrowski reads that peak as harmless carrier gas. Ferreira reads it as air leaking in. The two readings mean different repairs, on different days. Today you turn a pressure gauge into a number of molecules. You decide what one mass peak can and cannot prove on its own. Then you count each element's valence electrons, the outer ones that do the bonding. Eight hours of chamber time is the most expensive thing in the fortnight. It is being spent today.

BRIEFING 14W

The residual gas analyser has a peak that two people are reading two ways.

WHAT YOU DECIDE 15W

Count what is in a chamber from a gauge, and read a mass spectrum honestly.

4.1 How much is still in there

DEPOSITION BAY • A ROOM • BALLPARK

SCENE 28W

The chamber is at one millitorr with the wafer stage hot, and Vance wants the argument about whether that is empty settled in moles rather than in adjectives.

GUIDE 51W

Pressure, volume, amount and temperature are locked together, so three of them fix the fourth. Four tiles belong in that relation and two describe the chamber in ways that do not enter it. Pressure is in pascals and volume in cubic metres because the gas constant is quoted in those units.

ASK 12W

How many moles of gas are in the chamber at one millitorr?

TAKES AS READ

the gas constant relates pressure and volume to amount and temperature • absolute temperature is measured in kelvin

VERDICT 85W

A millionth of a mole sounds like nothing, and it is around six hundred million million molecules still in the chamber. A vacuum is a statement about how much has been removed rather than about emptiness. That number is what decides whether a precursor molecule crosses to the wafer untouched or reacts on the way. Note which quantities entered. The temperature matters because hot gas at the same pressure is fewer molecules, and the width of the chamber is already accounted for in its volume.

TAKEAWAY 24W

A pressure gauge and a chamber volume are a count of molecules, so a vacuum specification is a statement about how many are left.

4.2 Two gases, one mass

WAFER STORE • A PERSON • CHOICE

SCENE 35W

Twill has the residual gas trace pinned to the store noticeboard because it arrived with the lot paperwork. There is a peak at mass twenty-eight, and two people have written two different conclusions beside it.

GUIDE 47W

A mass spectrometer sorts what it is given by mass and charge, and reports a number rather than a name. The options differ in how much identification that number can carry on its own. What separates them is whether any other species could produce the same peak.

ASK 9W

What does a single peak at mass twenty-eight establish?

TAKES AS READ

a mass spectrum sorts particles by their mass and charge • a molecular mass is the sum of the masses of the atoms in the molecule

VERDICT 86W

Fourteen and fourteen is twenty-eight and so is twelve and sixteen, so nitrogen and carbon monoxide are indistinguishable at this resolution. One of them means air is getting in through a joint and the other means the carrier line is dirty, which are different repairs on different days. Separating them needs another peak: nitrogen fragments to mass fourteen and carbon monoxide gives twelve and sixteen, so the pattern rather than the single line is the identification. A spectrum names a mass; a pattern names a molecule.

TAKEAWAY 22W

A mass peak is a mass, and where two species share one, the peak alone cannot say which of them is there.

4.3 The outer shell, and nothing else

YELLOW BAY • A ROOM • PROTOCOL

SCENE 33W

Tanabe has four elements written on the board. They are the ones this layer is built from: silicon, boron, phosphorus and oxygen. She wants each outer shell counted before anybody draws a structure.

GUIDE 48W

Electrons fill shells in a known order and only the outermost ones take part in bonding. Match each element to what its outer shell holds and to what that lets it do. The count is a prediction about the number of bonds, made before any structure is drawn.

ASK 13W

Match each element to its outer shell and what that lets it do.

TAKES AS READ

electrons occupy shells in order, filling the inner ones first • the periodic table is arranged so that a group shares an outer-shell count

VERDICT 93W

The first two shells hold two and eight, so everything past ten is in the third shell and everything past two is in the second. Silicon's fourteen leaves four outside the filled ten; boron's five leaves three outside the filled two; phosphorus's fifteen leaves five; oxygen's eight leaves six. That count is the whole of what an atom brings to a bond. That is why elements in one column behave alike however different their masses are. It is also why the two columns either side of silicon are the ones this fab implants.

TAKEAWAY 19W

Only the outermost electrons take part in bonding, so counting them predicts how many bonds an atom will make.

4.4 The one number that names an element — Review

METROLOGY BAY • A ROOM • CALLBACK • CHOICE

SCENE 39W

Chidi Nkemdirim, the implant technician, has run boron into a test wafer and annealed it, and the travellers now call the disc doped silicon. Ferreira has been asked at the bench what on it is genuinely a second element.

GUIDE 53W

Four things the beam and the anneal did to that lattice, and they differ in which part of the atom they touched. Ask of each whether it changed the positive charge in a nucleus or only something outside one. Getting this wrong is how a damaged wafer gets reported as a contaminated one.

ASK 15W

After the implant, which of these has actually put a second element into the disc?

TAKES AS READ

an atom is a nucleus of protons and neutrons with electrons around it • an implanter fires ions into a solid hard enough to knock atoms out of place

VERDICT 84W

Only the proton count names an element, and only the boron nuclei carry a different one. A silicon atom knocked into a gap is silicon in the wrong place, which is a defect the anneal exists to repair. A silicon atom short of electrons is a silicon ion, the same element carrying a charge, and it takes electrons back as soon as any are available. A silicon atom vibrating harder is the same atom hotter. The lattice is disturbed four ways and renamed once.

TAKEAWAY 27W

A lattice can be displaced, ionised and heated without one atom in it changing what it is; only a nucleus with a different proton count does that.

END OF THE DAY 21W

A gauge counts molecules and a mass spectrum weighs them, and neither of them names what is there on its own.

The spectrum that is a picture of the shells

PLAN CARD 112W

Friday. The photoelectron spectrometer has the first failing wafer on it at nine. It is the one instrument here that answers what is on the surface and what it is bonded to.

Ferreira has waited eleven days for the slot. Ostrowski has asked for the chamber back by three. Vance cannot pump it down twice in one day. Today you turn a spectrum into binding energies. You predict how an element you have not measured should behave, from where it sits in the table. Then you work out how far a molecule travels in that chamber before it hits another one. Whatever the spectrum says decides which argument gets funded on Monday.

BRIEFING 11W

The surface measurement everybody has been waiting for runs this morning.

WHAT YOU DECIDE 15W

Read a photoelectron spectrum, and use the table to predict rather than to look up.

5.1 The photon in, the electron out

METROLOGY BAY • A ROOM • BALLPARK

SCENE 39W

Ferreira has the source at the aluminium line and a silicon peak on the screen. Dorsey wants the kinetic energy the instrument should see for that peak, before the run, so that a shifted peak is recognisable as one.

GUIDE 52W

The photon energy is known exactly and the binding energy is the quantity the peak identifies, so what the detector measures is whatever is left. Two binding energies are on the tiles: one belongs to silicon in the lattice and one to silicon bonded to oxygen. The question asks about the first.

ASK 9W

What kinetic energy should the silicon peak arrive at?

TAKES AS READ

a photon gives all of its energy to the electron it ejects • binding energy is the energy holding a particular electron to its atom

VERDICT 86W

The instrument cannot measure binding energy and does not try to. It measures how fast the ejected electrons are moving, and the difference between that and the known photon energy is what the atom was holding on with. Every peak in the spectrum is one shell of one element. Its position says how tightly that shell is held; its area says how many electrons are in it. That is why a single spectrum can carry an element list and a shell diagram at the same time.

TAKEAWAY 24W

The photon energy minus the measured kinetic energy is the binding energy, so a spectrum of peaks is a direct picture of the shells.

5.2 What the table says before the tool does

IMPLANT BAY • A PERSON • CHOICE

SCENE 36W

A supplier has offered gallium in place of boron for the shallow implant, on the grounds that both sit in the same column. Ganesan wants the prediction written down before anybody buys a cylinder of it.

GUIDE 47W

Two elements in one column share an outer-shell count and differ in how far out that shell sits. The options differ in which of those two facts they lean on. What decides between them is which properties follow from the count and which follow from the distance.

ASK 14W

What should be expected of gallium in place of boron, from its position alone?

TAKES AS READ

elements in one column of the table share the number of outer-shell electrons • a shell further from the nucleus is held less tightly

VERDICT 87W

A column is a statement about the outer-shell count. That count fixes how many bonds an atom makes, and so whether it is a donor or an acceptor in silicon. A row is a statement about how far out the shell sits and how much charge is shielded. That fixes the energies rather than the arithmetic. So gallium substitutes for boron in kind and not in degree. Three outer electrons either way, held more loosely, which shows up as a lower ionisation energy and a larger atom.

TAKEAWAY 29W

A column fixes what an element can do; a row fixes how strongly it does it. So a substitution down a group keeps the chemistry and changes the energies.

5.3 How far a molecule travels before it meets another

DEPOSITION BAY · A ROOM · BALLPARK

SCENE 38W

Ostrowski wants the pump-down shortened by four minutes and the chamber is thirty centimetres across. Vance will not agree to it until somebody says what the precursor is doing on the way to the wafer at each pressure.

GUIDE 58W

A molecule crosses the chamber freely when the average distance between collisions is larger than the chamber itself, and reacts in mid-flight when it is not. The tiles hold two pressures — the working pressure and the one outside — and only one of them is the question. The chamber width is there to compare the answer against.

ASK 12W

How far does a molecule travel between collisions at the working pressure?

TAKES AS READ

a lower pressure means fewer molecules in the same volume · molecules travel in straight lines between collisions

VERDICT 89W

Five centimetres between collisions in a chamber thirty centimetres across means most precursor molecules meet something on the way, and what they meet is usually each other. Reactions that happen in mid-flight produce particles that land on the wafer as defects rather than as film. Pump another decade down and the path is half a metre, longer than the chamber. So molecules travel from the inlet to the wafer untouched, and the only place chemistry happens is the hot surface. That is the whole argument for the pump-down time.

TAKEAWAY 27W

The average distance between collisions is set by the pressure, and it decides whether a process is an arrival at a surface or a reaction in mid-flight.

5.4 Why the bay is lit amber — Review

YELLOW BAY · A ROOM · CALLBACK · SEQUENCE

SCENE 44W

Park has four light sources listed for the bay refit — a new exposure lamp, the amber safelight, an inspection illuminator and an alignment laser. Tanabe will not sign the order until they are ranked by what each one can do to a resist.

GUIDE 55W

Four sources, all of them switched on in the same room, so nothing here happens before anything else. What orders them is the energy carried by one packet of each, and the constants in that expression are the same for every lamp in the building. The watts on the labels are a rate of arrival.

ASK 12W

Order the four sources by the energy one photon of each carries.

TAKES AS READ

light of a given wavelength arrives in packets of a fixed energy · Planck's constant and the speed of light do not vary between lamps

VERDICT 83W

The energy one packet carries is fixed by the wavelength alone, so the ranking is the wavelength order reversed and the wattage never enters it. The alignment laser at 193 nanometres carries three times what the amber safelight does. That is why eight tenths of a watt of it will expose a resist and thirty watts of amber will not.

Brightness decides how many packets arrive each second; the wavelength decides whether a single one of them can do the job at all.

TAKEAWAY 24W

Ranking light sources by what they can do to a material ranks them by wavelength, and the power rating plays no part in it.

END OF THE DAY 25W

A photon of known energy in and an electron of measured energy out is a direct reading of how tightly an atom was holding on.

Four numbers that agree, and one that does not

PLAN CARD 113W

Monday. The second week opens with four measurements of the substrate that look independent. They all agree to within a few per cent. A yield map says two of the lots are not the same material. Twill has the certificates. Ferreira has the mapped results. Hale has told them both that agreement is not the same as evidence. Today you open those four numbers and find what each of them was measured against. You turn a purity limit into a count of atoms. Then you draw the first molecule this process is built out of. If the substrate is wrong, every recipe argument of the last three weeks was about the wrong thing.

BRIEFING 11W

Every incoming measurement is consistent, and the lots are behaving differently.

WHAT YOU DECIDE 14W

Find out why every measurement of the substrate agrees and the wafers still fail.

6.1 What they were all measured against

WAFER STORE · A ROOM · TRACE

SCENE 35W

Four substrate results, agreeing closely, and two lots behaving differently on the floor. Twill has the retained boxes and the certificates, and Ferreira has asked what each of the four numbers was actually compared with.

GUIDE 45W

Open each channel to see what it was derived from. Keep the ones whose evidence still stands on its own, then name the source everything else went through. A number compared against a reference inherits whatever is wrong with that reference, and inherits it quietly.

ASK 22W

Everything except the hot-probe reading agrees. Open the derivations, keep what still stands on its own, and name the source at fault.

BEHIND THE BUTTON 154W

Why agreement is weak evidence. Four quantities computed against one reference move together whatever that reference is doing, so their agreement is a property of the arithmetic rather than of the material. It is the most convincing form a systematic error takes, because nothing anywhere looks inconsistent.

What a resistivity measurement can and cannot say. Resistivity is a magnitude. A substrate doped with boron and one doped with phosphorus can have the same resistivity to within a per cent. Resistivity does not depend on which of the two columns the dopant came from. Separating them needs a measurement that responds to the sign of the carriers rather than to how many there are.

What the certificate has no field for. It records resistivity, thickness, flatness and particle count. Type is not on it, because for thirty years this supplier has shipped one type and nobody wrote a field for a thing that never varied.

TAKES AS READ

a quantity computed against a reference carries the reference's error · resistivity measures how much conduction there is, not what is carrying it

VERDICT 100W

Three of these numbers are comparisons, and a comparison carries whatever is wrong with what it was compared against. The reference wafer was assumed to be one type on the strength of a resistivity band, and resistivity is a magnitude that both types can have equally. So three measurements agree beautifully while all describing the same wrong assumption, which is exactly what makes a common reference error survive review. The hot-probe test responds to the sign of the carriers rather than to their number. The sealed box has been through no measurement at all. So neither of them should move.

TAKEAWAY 18W

A specification records what somebody thought to measure, so two materials can meet it identically and behave differently.

6.2 Five in a billion, counted

IMPLANT BAY • A ROOM • BALLPARK

SCENE 38W

Ganesan has the certificate limit on the board beside the implant recipe, which is quoted in atoms per cubic centimetre. Nkemdirim, the implant technician, has asked which of the two numbers is the bigger intervention in the lattice.

GUIDE 49W

A ratio becomes a count once you know how many atoms are there to begin with. Two tiles make the ratio into a fraction and one says how many silicon atoms sit in a cubic centimetre. The lot size is on the tiles and belongs to a different question.

ASK 12W

How many boron atoms per cubic centimetre does the incoming limit permit?

TAKES AS READ

a part per billion is one atom in a thousand million • the silicon lattice has a fixed number of atoms in a given volume

VERDICT 86W

A ratio that sounds like nothing is a very large number of atoms, and the lattice responds to the number. Two hundred and fifty million million boron atoms sit in every cubic centimetre. That is within a factor of a few of the dose this bay implants on purpose, which is why the incoming limit is set where it is. It is also why doping works at all. An atom from the next column brings one electron too few to a lattice that needs four bonds.

TAKEAWAY 29W

A few parts per billion of an atom from the neighbouring column is millions of millions of atoms per cubic centimetre. That is why the impurity is the product.

6.3 Where the electrons are in a precursor

DEPOSITION BAY • A PERSON • PROTOCOL

SCENE 28W

Ostrowski has four precursor molecules on the whiteboard with no electrons drawn on any of them. The recipe argument keeps returning to how easily each one comes apart.

GUIDE 43W

A Lewis structure spends every valence electron the atoms bring, as bonding pairs or as lone pairs. Match each molecule to what its structure has to contain. Counting first and drawing second is what stops a structure being invented to fit an expectation.

ASK 10W

Match each precursor to what its Lewis structure must contain.

TAKES AS READ

each atom contributes its outer-shell electrons to the molecule • a shared pair between two atoms is a single bond

VERDICT 95W

The arithmetic is the same every time. Add up the outer-shell electrons the atoms bring, then use two for each bond. Put whatever is left on the atoms as lone pairs until each one has eight. Silicon brings four and four hydrogens bring one each, which is exactly four bonds and nothing over. Nitrogen brings five, uses three, and keeps a pair. Oxygen brings six, uses two, and keeps two pairs. Chlorine brings seven, uses one, and keeps three — which is why the two chlorines dominate the electron count in the last of them.

TAKEAWAY 21W

A Lewis structure spends every valence electron the atoms bring, as bonds or as lone pairs, and nothing is left over.

6.4 The top nanometre, and everything under it — Review

WET BENCH • A ROOM • CALLBACK • PROTOCOL

SCENE 40W

Ivy Bramall, the wet bench operator, has four results back on a single rinsed wafer and two of them disagree. Aubrey will not change the bath chemistry until each result has been attributed to what it was actually looking at.

GUIDE 54W

Four methods and four statements, and the pairing turns on how far into the wafer each one reached rather than on how good the instrument is. Ask of each what quantity of silicon its number was divided by. Every statement is used once, so settling the two you are sure of constrains the rest.

ASK 13W

Match each result on the wafer to what it is a statement about.

TAKES AS READ

a contamination figure is an average over whatever the instrument sampled • a wafer is about three-quarters of a millimetre thick and a device layer a few nanometres

VERDICT 81W

Every method reports a total divided by whatever it sampled, so the depth it reached is part of the answer rather than a detail of the instrument. Dissolving the disc divides a surface contaminant by three-quarters of a millimetre of silicon and loses it in the average. The spectrum divides it by a nanometre and finds it, and says what it is bonded to. The area count places it without depth. Only the profile answers depth, and it costs the wafer.

TAKEAWAY 21W

How deep a method reached is part of what its number means, so a contamination result cannot be read without it.

END OF THE DAY 15W

Numbers computed against a common reference agree with each other whatever that reference is doing.

Shapes, shifts and a sea of electrons

PLAN CARD 116W

Tuesday. The substrate finding has been on Hale's desk since eight. Two lots go back to the supplier. Thirteen do not. The failures on those thirteen are still unexplained, and eight days are left to explain them. Ostrowski has taken the finding as proof he was right all along. He wants the recipe review to start this afternoon. Today you read a peak in a spectrum for what it says about bonding, not about how much is there. You work out what shape each precursor molecule is forced into. Then you name what actually holds a metal together. The recipe review cannot tell a wrong film from a merely different one until all three are settled.

BRIEFING 10W

The substrate finding has changed which questions are worth asking.

WHAT YOU DECIDE 15W

Read what a peak position says about bonding, and name what holds each material together.

7.1 The same element, four electronvolts over

METROLOGY BAY • A ROOM • CHOICE

SCENE 33W

Dorsey's overnight run has two silicon peaks on one wafer, four electronvolts apart, and the summary report lists silicon once. Ferreira wants the difference between the two explained before anybody quotes a quantity.

GUIDE 56W

A peak position is a binding energy and a binding energy is how tightly an electron is held. The options differ in whether they read the split as a difference in amount, in instrument behaviour or in bonding. What separates them is what could move a peak without changing how much of the element is present.

ASK 9W

What do two silicon peaks four electronvolts apart establish?

TAKES AS READ

a peak position is a binding energy and a peak area is a quantity • an electronegative neighbour draws electron density away from an atom

VERDICT 92W

Oxygen pulls electron density away from the silicon it is bonded to. That leaves the remaining electrons on that silicon feeling more of the nucleus, and held more tightly. More tightly held means a higher binding energy, which is a peak a few electronvolts over from the bare lattice. So the split is a statement about neighbours. The two peaks are the same shell of the same element in two chemical situations. Their relative areas say how much of the surface is oxide. That is the measurement no bulk method can make.

TAKEAWAY 22W

A shift in peak position says what the atom is bonded to; only the area says how much of it there is.

7.2 What the lone pairs do to the shape

DEPOSITION BAY • A PERSON • PROTOCOL

SCENE 36W

Ostrowski has the four precursor structures drawn up from yesterday. He has also been asked the same question twice this week. Why do two molecules with the same number of atoms behave differently at one temperature?

GUIDE 55W

Count the domains around the central atom — every bond and every lone pair — and let them get as far apart as they can. Match each molecule to the shape that count forces. Lone pairs occupy space and take no atom with them, which is why they change the shape without appearing in it.

ASK 10W

Match each molecule to the shape its electron domains force.

TAKES AS READ

electron domains repel one another and settle as far apart as they can • a lone pair occupies a region around the atom just as a bond does

VERDICT 99W

The rule is one count and one principle. Every bond and every lone pair is a region of negative charge, and regions of negative charge get as far from each other as they can. Four regions settle at about a hundred and nine degrees whether or not each one carries an atom. So ammonia and water are built on the same arrangement as silane, and simply have some corners empty. That is why the angle closes as lone pairs are added. A lone pair is held by one nucleus rather than two, spreads wider, and pushes the bonds together.

TAKEAWAY 20W

A shape follows from a count of electron domains, and lone pairs bend a molecule without appearing in its outline.

7.3 Electrons that belong to none of them

WET BENCH · A ROOM · CHOICE

SCENE 35W

Aubrey has a copper fill step that has to conduct and survive being polished flat. Bramall, the wet bench operator, has asked why the copper deforms under the pad while the oxide beside it chips.

GUIDE 44W

The options differ in where they put the electrons in a metal. What separates them is having to explain two facts at once with the same picture. A metal conducts, and it deforms rather than shattering when its atoms are pushed past each other.

ASK 13W

What explains both that copper conducts and that it deforms under the pad?

TAKES AS READ

a metal atom holds its outer electrons loosely · conduction requires charge that is free to move

VERDICT 96W

A pool of electrons belonging to no particular pair of atoms explains both observations at once. Charge that is not tied down can move, which is conduction. And because no bond points from one specific atom to another, a plane of atoms can slide over its neighbour. It is held just as well in the new position, which is deformation rather than fracture. An ionic solid does the opposite. Slide a plane and like charges come to face each other, so it splits along that plane. That is what the oxide beside the copper is doing.

TAKEAWAY 25W

A metal is a lattice of cations in a shared pool of electrons, which is why it conducts and why it deforms instead of shattering.

END OF THE DAY 20W

A peak that has moved and a molecule that is bent are both statements about bonding rather than about amount.

Which structure is right, and what it costs to pull a solid apart

BEFORE THE DAY — HUNT

Six retained boxes that settle the substrate argument

Twill keeps a sealed sample of every incoming lot, and those boxes are the only material in this building that has never been through a tool. Nine of them are out on the floor and the log accounts for six. Six will settle the substrate argument, so stop at six; a box nobody can find is either evidence or a box somebody has already opened.

PLAN CARD 111W

Wednesday. The recipe review has stalled on a molecule that can be drawn two ways. Vance needs a purchase decision on the precursor by Friday. The two drawings mean different handling, different lines and a different temperature. Meanwhile Twill has three ceramic liners quoted at three prices. All three are sold as equivalent. Today you settle which of the two drawings is the better description. You work out whether the molecules on the board are polar. Then you rank three solids by what it takes to pull them apart. Buying the wrong precursor would cost eleven days. There are only fifteen in the fortnight, so that is the rest of it.

BRIEFING 10W

Two structures for the same molecule imply two different processes.

WHAT YOU DECIDE 14W

Choose between structures that both satisfy the octet, and rank what holds solids together.

8.1 Two drawings, one molecule

DEPOSITION BAY • A ROOM • CASEBOOK

SCENE 32W

Two structures for the same precursor are on the board, both giving every atom a full outer shell. Vance needs one of them chosen before the purchase order goes out on Friday.

GUIDE 51W

Formal charge counts how far each atom sits from the electron count it brought. Match each drawn structure to what its charges say about it. A structure with charges spread onto atoms that do not want them is a worse description of a real molecule, even where every octet is satisfied.

ASK 10W

Match each proposed structure to what its formal charges say.

TAKES AS READ

each atom in a structure can be assigned a share of the electrons around it • a full outer shell can often be achieved in more than one arrangement

VERDICT 90W

Formal charge is bookkeeping, not a real charge. It asks what each atom would be holding if every bond were split down the middle, and compares that with what the atom brought. Small numbers mean the electrons are distributed roughly where each atom expects them, and large numbers mean the arrangement is straining. The one rule that decides most cases is that negative charge should sit on the atom that pulls hardest. So a large negative on the least electronegative atom is a structure to redraw rather than to defend.

TAKEAWAY 22W

Where two structures both fill the octet, the one with the smaller formal charges is the better description of the real molecule.

8.2 Shape decides what the pulls add up to

YELLOW BAY • A PERSON • CHOICE

SCENE 37W

Tanabe has two solvents on the bench that share the same kinds of bond and behave nothing alike on the wafer. Park wants to know how to tell them apart before he loads either into the track.

GUIDE 50W

A bond is polar when the two atoms pull unequally, and a molecule is polar only when its shape leaves those pulls uncanceled. The options differ in which of the two conditions they treat as sufficient. The test is whether a molecule of polar bonds can come out non-polar overall.

ASK 11W

What decides whether a molecule with polar bonds is itself polar?

TAKES AS READ

electronegativity is how strongly an atom pulls shared electrons towards itself • a molecule has a definite three-dimensional shape

VERDICT 93W

Each polar bond pulls in a direction, and directions add up. Four identical bonds arranged symmetrically around a central atom pull equally outward in every direction and leave nothing over. So the molecule is non-polar, however polar each bond is. Take one of those bonds away and replace it with a lone pair, and the balance is broken and the molecule has a direction. That is why carbon dioxide is non-polar and water is not, although both are built entirely of strongly polar bonds. It is also why the two solvents behave differently.

TAKEAWAY 24W

Two molecules built out of the same bonds can behave completely differently, and what separates them is how those bonds are placed in space.

8.3 Charges, spacing, and a melting point

WAFER STORE · A ROOM · SEQUENCE

SCENE 30W

Twill has three ceramic liners quoted as equivalent at three prices, and the highest process temperature is not yet fixed. He wants them ranked before he signs for the cheapest.

GUIDE 49W

The energy holding an ionic solid together grows with the charges on the ions and falls as the ions get further apart. Order the four by that energy, using the charges and the spacings given. Charge does more work than spacing here, which is what the ordering is for.

ASK 15W

Order the four ionic solids by how much energy it takes to pull them apart.

TAKES AS READ

an ionic solid is a lattice of oppositely charged ions · attraction rises with charge and falls with separation

VERDICT 91W

Both terms in the attraction are visible here. Closing the spacing from a third of a nanometre to under a quarter raises the energy by roughly half. Doubling one of the charges doubles it outright, and doubling both quadruples it. That is why charge sorts this list and spacing only fine-tunes it. It is also why the liner made of doubly charged ions holds together to a temperature the others do not reach. The same arithmetic explains why the melting points of these four run over more than a thousand degrees.

TAKEAWAY 24W

Lattice energy rises with the charges on the ions and falls as they get further apart, and charge is by far the stronger term.

END OF THE DAY 27W

A structure that satisfies the octet may still be the worse description, and what a solid costs to take apart follows from its charges and their spacing.

The ledger of breaking and making

PLAN CARD 115W

Thursday. The film analysis came back overnight. It shows the composition Ostrowski predicted three weeks ago, when nobody would pay for the measurement. The nitride on the failing lots has the right two elements in the wrong proportion. The furnace log adds one more fact. The deposition temperature has been running eighteen degrees below its setpoint since a controller was replaced in February. Today you work out how much energy it takes to break the precursor apart. You compute what the film should be made of if the recipe were followed exactly. Then you settle what holds one molecule to the next. This is the day Ostrowski turns out to have been right as well.

BRIEFING 12W

The nitride film is the right two elements in the wrong proportion.

WHAT YOU DECIDE 18W

Establish what a temperature has to be high enough to break, and what the film is made of.

9.1 What breaking the precursor costs

DEPOSITION BAY · A ROOM · BALLPARK

SCENE 33W

Ostrowski has the furnace log open at the February controller change and the precursor structure beside it. Nobody has yet costed what the reaction has to pay for and what it gets back.

GUIDE 53W

Breaking bonds costs energy and making them returns it, so the net is one sum minus the other. Four tiles are the two bond enthalpies and how many of each there are; the fifth belongs to a bond that appears nowhere in this reaction. Count the bonds in the drawing rather than guessing.

ASK 12W

What is the net energy of decomposing silane into silicon and hydrogen?

TAKES AS READ

a bond enthalpy is the energy needed to break one mole of that bond · the atoms are conserved, so every bond broken is replaced by bonds made

VERDICT 80W

Four silicon-hydrogen bonds have to be broken. Two hydrogen-hydrogen bonds are made in their place, and the difference is what the furnace pays. It is positive, so the reaction takes energy in. It will not run without a hot surface to supply that energy, which is why a deposition temperature is not a preference. A precursor built with stronger bonds needs more. That is why swapping silane for a chlorinated molecule raises the temperature the recipe has to run at.

TAKEAWAY 27W

A reaction is the difference between the energy of the bonds broken and the bonds made, which is why one precursor needs a hotter wafer than another.

9.2 The formula the recipe claims

WAFER STORE • A PERSON • BALLPARK

SCENE 33W

The analysis of the failing film reads sixty-six per cent silicon by mass. Twill has the specification for the layer pinned beside it and nobody has worked out what the specification actually implies.

GUIDE 48W

A formula is a claim about proportions, and it becomes a number through molar masses. Three tiles carry the silicon side of the formula unit and the mass of the whole. The other two describe the nitrogen side, and belong to the other half of the same calculation.

ASK 9W

What percentage of silicon nitride is silicon, by mass?

TAKES AS READ

a chemical formula gives the ratio of atoms in a compound • a molar mass is the sum of the molar masses of the atoms in the formula

VERDICT 81W

The specified film is sixty per cent silicon by mass. The measured film is six points above that, which is not a small discrepancy but a different material. The extra silicon has nowhere to sit except as silicon that never found nitrogen to bond to. That changes what the layer does electrically, which is the failure the yield map shows. It also shows why composition is the right measurement here. A layer can be the correct thickness and the wrong substance.

TAKEAWAY 22W

A formula asserts a proportion by mass, so a measured composition either matches it or names how far the film has drifted.

9.3 The forces between whole molecules

YELLOW BAY · A ROOM · CHOICE

SCENE 29W

Two developer solvents boil ninety degrees apart and their molecules are almost the same size. Tanabe wants the reason stated before the second one is qualified for the track.

GUIDE 51W

The forces between separate molecules are far weaker than the bonds inside them, and they come in a ranked family. The options differ in which force they attribute the boiling point to. What separates them is whether the property being explained belongs to the molecule or to the attraction between molecules.

ASK 13W

Why do two solvents of almost the same size boil ninety degrees apart?

TAKES AS READ

boiling separates molecules from one another without breaking bonds inside them · a polar molecule has a permanent positive and negative end

VERDICT 81W

Boiling pulls molecules away from each other and leaves every molecule intact. So what a boiling point measures is the attraction between them. That attraction runs in a ranked family. Dispersion is in everything, dipole attraction where the molecule is polar, and hydrogen bonding where hydrogen reaches a lone pair on the next molecule. The last of those is several times the strength of the first. That is easily enough to account for ninety degrees between two molecules of similar size.

TAKEAWAY 26W

Boiling separates molecules from each other rather than breaking them, so a boiling point measures the attraction between molecules and not the strength of their bonds.

END OF THE DAY 26W

A temperature has a job to do, and a film with the right elements in the wrong ratio is a temperature that did not do it.

What each material is held together by

PLAN CARD 111W

Friday. Two causes are established, with five working days spent to get them. The substrate was not what everybody took it to be. The furnace has been running cold. Hale wants the layer stack itself understood before the qualification recipe is written. A stack is four materials, and the same polish treats all four differently. Vance has a moisture reading somewhere in six hundred metres of gas line. She has one shift to find it. Today you name what holds each material in the stack together. You settle what makes silicon switchable at all. Then you walk the gas line until the pattern breaks. Five days are left after this one.

BRIEFING 14W

Two causes are established and the layer stack has to be understood as materials.

WHAT YOU DECIDE 15W

Name what holds each of these materials together, and find where the gas line breaks.

10.1 Four materials, four things holding them

WAFER STORE • A ROOM • CHOICE

SCENE 39W

Twill has four samples from the stack on the bench. They are the silicon substrate, the oxide, the copper fill and a block of resist. One polish step treats all four the same way and gets four different results.

GUIDE 40W

The four options are four different things a solid can be held together by. What separates them is which one explains a melting point above fourteen hundred degrees in a material that carries no charge and contains only one element.

ASK 13W

What holds the silicon substrate together, and why does it melt so high?

TAKES AS READ

a melting point measures what has to be overcome to separate a solid • a covalent bond between two atoms is far stronger than an attraction between molecules

VERDICT 95W

There are no silicon molecules. Every atom is bonded to four neighbours, and those to four more. So the whole crystal is one connected structure, and there is no small unit to pull away from the rest. Melting it therefore costs what a covalent bond costs, several hundred kilojoules per mole, rather than what an attraction between molecules costs, which is a few tens. The resist on the same bench is the other extreme: real molecules, held to each other weakly. That is why one softens in a warm oven and the other does not.

TAKEAWAY 27W

What has to be overcome to melt a solid is whatever is holding it together, and that is a different thing in each of these four materials.

10.2 The gap, and what can clear it

IMPLANT BAY • A PERSON • SEQUENCE

SCENE 35W

Ganesan has an undoped wafer that conducts almost nothing at room temperature, conducts when it is warmed, and conducts when a lamp is shone on it. Nkemdirim wants the three observations explained with one picture.

GUIDE 50W

In a covalent network every electron is committed to a bond, and freeing one costs a fixed amount of energy. Order the four situations by how much energy is available to free an electron. The point of the ordering is that three different supplies of energy do the same job.

ASK 10W

Order the four situations by how much conduction each produces.

TAKES AS READ

conduction requires electrons that are free to move • an electron in a covalent bond is held by two nuclei at once

VERDICT 96W

Every electron in pure silicon is committed to one of four bonds, and getting one out costs a definite amount of energy. At room temperature almost nothing has that much, so almost nothing conducts. Warmth supplies it to a small fraction. Light supplies it in single packets to whichever electrons the photons meet, which is why the threshold is a wavelength rather than a brightness. A dopant does something different and far more effective. It brings an electron that was never in a bond to begin with, so no energy has to be found at all.

TAKEAWAY 26W

Freeing an electron in a covalent network costs a fixed amount of energy, and light, warmth or a dopant atom are three ways of supplying it.

10.3 Six hundred metres of line, one reading

DEPOSITION BAY • A ROOM • PROBE

SCENE 37W

The chamber sees moisture that the cylinder certificate says is not in the gas. Vance has six stations down the delivery line, each with a tap and an analyser port, and one shift before the weekend shutdown.

GUIDE 56W

Take a reading at whatever station you want. Every reading costs time and none of them costs anything else. What you are looking for is the first station where the number departs from what it should be. Everything downstream of an entry point carries the same contamination, so the deepest bad reading is not the answer.

ASK 23W

The chamber sees moisture the cylinder does not contain. Take the readings you think you need — where is it entering the line?

TAKES AS READ

a contaminant entering a line is carried downstream of where it entered • a gas analyser reports what is in the stream at the point it is tapped

VERDICT 102W

Three stations held exactly what they should, which is what rules out the cylinder, the regulator and the purifier together. A bad cylinder would show at the outlet. A failed purifier would show as no improvement rather than as an improvement followed by a rise. From the manifold onwards every station is high, and a contaminant that enters a flowing line is carried downstream and not upstream. So the worst reading is not the fault: the fault is the first departure, and everything below it is inheriting. The manifold was rebuilt in February, which is the same month the controller was changed.

TAKEAWAY 24W

In a chain, the fault is at the first station that departs from expectation, and every station after it is inheriting the same problem.

END OF THE DAY 15W

What a material does when heated, polished or wetted follows from what holds it together.

A clean surface is an event, not a state

PLAN CARD 108W

Monday, with four working days left. Aubrey has wafers that leave her bench bare and reach the coater with an oxide on them. The process does not allow that oxide. There are four candidate causes between the two rooms. Park has the queue times for every wafer for six weeks. Nobody has yet compared them against the failures. Today you find out which of the four steps is putting the oxide back. You decide which solvent will lift the resist and leave the layer underneath it. Then you read a spectrum with two things wrong with it at once. A wafer that waits is a wafer that changes.

BRIEFING 16W

The bare surface the next step needs is not bare by the time it gets there.

WHAT YOU DECIDE 16W

Find out what is putting oxide back on the wafer, and what a solvent will lift.

11.1 What is regrowing on the wafer

WET BENCH · A ROOM · CONTROL

SCENE 31W

Wafers leave Bramall's bench bare and reach the coater with oxide on them. Four things changed between the two rooms in February and the log makes all four look equally guilty.

GUIDE 62W

The number to watch is the oxide signal as a percentage of the silicon signal on the incoming wafer. Four things changed and that is four suspects and no evidence. Change one, run the measurement, put it back, and name the one the oxide follows. A change that moves two things at once answers a vaguer question than the one being asked.

ASK 16W

Four things changed and one of them is regrowing the oxide. Establish which, and prove it.

BEHIND THE BUTTON 135W

Why the log is a list of suspects. Four changes in one month is ordinary in a working fab, and the oxide had to start somewhere. A record of what changed says where to look, and all four would read as equally plausible in a report.

What a bare silicon surface actually is. Cutting or etching a covalent network leaves atoms at the surface with bonds that reach nothing. Oxygen and water satisfy them within minutes. That is why the native oxide is not contamination in the ordinary sense: it is the surface reaching the state it prefers.

Why the reversal matters. A cause is something the effect depends on in both directions. Putting the change back and watching the oxide return rules out the ordinary alternative: that something else was drifting while you worked.

TAKES AS READ

a controlled test changes one variable and holds everything else · a bare surface has bonds that are not satisfied by anything

VERDICT 94W

A cleaned silicon surface is not a stable thing. Etching a covalent network leaves surface atoms with bonds reaching nothing, and oxygen and water satisfy them in minutes. So what looks like contamination is the surface going to the state it prefers. That makes exposure time the variable and everything else in the room secondary. The reversal is what turns a candidate into a cause, because a coincidence does not come back on demand. Three of these four changed in the same month as the oxide appeared without having anything to do with it.

TAKEAWAY 23W

A relationship is causal when reversing it brings the effect back, not when the two happen to have changed in the same month.

11.2 What the developer will lift

YELLOW BAY · A ROOM · CHOICE

SCENE 28W

Tanabe has two developers qualified and a resist whose exposed regions become polar while the unexposed regions stay non-polar. Park has to choose one before the afternoon lot.

GUIDE 51W

A solvent dissolves what it resembles, because replacing one set of attractions between molecules with a similar set costs almost nothing. The options differ in what they treat as the deciding property. The distinction decides which half of the pattern is removed and which half survives to protect the layer beneath.

ASK 11W

Which developer removes the exposed resist and leaves the rest standing?

TAKES AS READ

dissolving replaces attractions between like molecules with attractions between unlike ones · exposure changes the resist chemically rather than removing it

VERDICT 96W

Dissolving is an exchange: a molecule leaves the attractions it had with its own kind and takes up attractions with the solvent instead. Where the two kinds of attraction are alike, the exchange is roughly free and the material dissolves. Where they are not, it costs energy and the material stays put. So a polar developer takes the exposed polar regions and leaves the unexposed non-polar ones standing, which is exactly the selectivity the pattern depends on. Volatility decides how fast the developer leaves afterwards, which is a different question from what it takes with it.

TAKEAWAY 26W

Dissolving is an exchange of attractions, so what a liquid takes off a surface follows from what it resembles rather than from how strong it is.

11.3 One spectrum, two findings

METROLOGY BAY • A PERSON • DIAGNOSIS

SCENE 31W

Ferreira has the incoming wafer on the spectrometer with the queue time deliberately left long. The spectrum has an oxide peak that everybody expected and a second feature that nobody did.

GUIDE 56W

Four panels are on the screen and only one explanation accounts for all four at once. The quiet readings are what rule the others out. Read the ones that look normal before the one that is alarming. An explanation that accounts only for the loud panel is the one that gets accepted, and it is wrong.

ASK 10W

Which explanation fits every panel, not just the loudest one?

TAKES AS READ

a shifted peak marks a chemical state and a new peak marks a new element • an argon-ion etch removes the top layers of a surface

VERDICT 99W

Two of these panels are quiet and they are what settle it. The reference sample is clean in the carbon region, so the instrument is not the source; if it were, everything it measured would carry the same peak. And the argon etch removes the oxide and leaves the carbon. That puts the carbon underneath rather than on top. An adsorbed layer picked up in the chamber would come off first. So there are two findings on one spectrum, and only the second of them is new: the oxide is the queue time, and the carbon arrived before it.

TAKEAWAY 25W

An explanation has to account for the quiet readings as well as the loud one, and the quiet ones are what rule the alternatives out.

END OF THE DAY 22W

A freshly cleaned surface has unsatisfied bonds and the air satisfies them, so every step after a clean is racing a clock.

Inside the limits, and worth knowing

PLAN CARD 102W

Tuesday. The qualification recipe has to be frozen on Thursday morning. Ganesan's implanter is delivering a dose whose spread is wider than the customer allows. She can re-aim the beam. She can also measure the dose better. Those are not the same choice. Ferreira has six hours of spectrometer time left in the fortnight and four proposals for it. Aubrey wants to know whether the copper is alloyed or pure before the fill step is written. Today you settle all three of those questions. Hale is signing whatever comes out of today. What is not decided today is not in the lot.

BRIEFING 15W

Three days left and more things worth knowing than there is time to know them.

WHAT YOU DECIDE 14W

Bring the implant dose inside specification, and buy only the measurements that decide something.

12.1 Aiming it and knowing it are different

IMPLANT BAY · A ROOM · CLOUD

SCENE 35W

Ganesan has the implanted dose from four hundred wafers plotted against the customer limits. The cloud sits off centre and it is also wider than the band, and she has six hours of tool time.

GUIDE 60W

The cloud is every wafer the tool has delivered, and the band is what the customer will accept. Re-aiming slides the whole cloud sideways; a measurement pass narrows it. When you are ready, place the middle bar where you say the dose is centred. Put the two outer bars a single standard deviation either side of it, then declare it.

ASK 25W

Bring the delivered dose inside the customer band, place the bars to report the middle and its one-sigma spread, and say when it is ready.

BEHIND THE BUTTON 143W

Why the spread is not the aim. Two different things are wrong with this distribution. One is where its middle sits, which a re-aim fixes. The other is how wide it is, which is a statement about how well the dose is known and cannot be improved by aiming better.

What a standard deviation means here. Placed correctly, the two outer bars hold about two thirds of the wafers between them, and the middle bar has about half the points on either side. Those are properties of the cloud on the screen rather than of the specification.

Why a dose has a spread at all. The implanter counts charge and the lattice receives atoms, and between the two are a beam that wanders, a wafer that heats and a monitor that samples. Each of those is small and none of them is zero.

TAKES AS READ

a dose measurement is a distribution rather than a single value · a specification band is a limit on individual wafers rather than on their average

VERDICT 101W

The middle of this distribution is the one dose the tool is least likely to deliver on any given wafer. Sliding it to the centre of the band helps, and it does not help enough. The wafers falling outside are doing so on the width rather than on the offset. Move the middle and the ones that were high become low instead. Only a better measurement of each wafer's own dose reduces that width, and it costs tool time the day would rather spend elsewhere. That is the real decision, and the band is a customer limit that does not negotiate.

TAKEAWAY 16W

Re-centring moves a whole distribution without changing its width, and only new measurement makes it narrower.

12.2 Six hours, four proposals

METROLOGY BAY • A ROOM • VALUE

SCENE 32W

Ferreira has six hours of spectrometer time left before Thursday and four things worth measuring. Hale has said she will fund whatever changes what the fab does and nothing that does not.

GUIDE 52W

Every one of these four is real evidence and would be worth having in a quieter month. Open each and ask what the fab would do differently on Thursday depending on how it comes out. A measurement whose result changes nothing before the recipe freezes has spent the hours on a report.

ASK 20W

You have six hours of spectrometer time before the recipe freezes. Buy what changes what the fab does on Thursday.

BEHIND THE BUTTON 97W

What makes a measurement decisive. Two possible results that lead to two different actions. Hale will not say which she expects, because the expectation is not the point — the question is which result could change the recipe.

Why Thursday is the constraint. The recipe freezes on Thursday morning whatever anybody knows by then, so information arriving afterwards is information about a decision already taken.

Why all four are good proposals. A choice between one good measurement and three poor ones tests nothing. The skill is separating valuable from decisive, and they are not the same property.

TAKES AS READ

evidence is worth what it changes rather than what it costs • a second measurement of the same kind narrows the same thing again

VERDICT 95W

Two of these proposals measure the film that has already failed. They are careful, and they will produce a better number about a wafer nobody is going to ship. That is the failure this decision exists to avoid. A measurement can be rigorous and still answer a question whose answer does not enter the choice. The December wafers would confirm the history and change no recipe. The carbon profile is a genuine second finding and it belongs to next month. Only the overnight wafer can come back the other way and stop the temperature change.

TAKEAWAY 15W

A measurement earns its cost by changing a decision that has not yet been taken.

12.3 A second element in the metal

WET BENCH • A PERSON • CHOICE

SCENE 36W

Aubrey has two copper sources quoted for the fill: one pure, one carrying a per cent of a second element. The polish step pushes hard on whatever goes in and the fill has to survive it.

GUIDE 52W

The options differ in what a small addition to a metal does to it. What separates them is whether the change is about the electron pool or about the planes of atoms sliding over one another. A per cent by mass is a great many atoms and almost no change in composition.

ASK 14W

What does one per cent of a second element do to the copper fill?

TAKES AS READ

a metal deforms when planes of atoms slide over each other • a metal conducts through electrons that are not tied to particular atoms

VERDICT 87W

A metal deforms because planes of atoms can slide and be held just as well in the new position. Anything that interrupts a plane makes that harder. An atom of a different size, sitting in place of a copper atom or squeezed into a gap, does exactly that. It does it at every one of the enormous number of sites one per cent provides. The electron pool is barely touched, so the alloy still conducts. That is why this is the standard way to strengthen a metal.

TAKEAWAY 29W

A metal gives way where its atoms can move past each other. So a very small amount of something in the way is enough to change how it behaves.

END OF THE DAY 22W

Re-aiming moves a distribution and only new information narrows it, and a measurement is worth what the decision it changes is worth.

Which step decides whether the film grows

BEFORE THE DAY — CANVASS

Who has actually seen the February controller change

The furnace controller was replaced in February and the work order names one engineer who has since left. Ask round until enough people have said what they actually saw rather than what the work order says, because a fab this size runs on twelve memories and one of them becomes a fact if nobody checks it.

PLAN CARD 103W

Wednesday. The recipe goes to Hale tomorrow morning. Ostrowski has five steps between a gas cylinder and a finished film. He wants to know which of the five decides the outcome. That is the step the recipe has to guarantee. Twill has money for two more pieces of work and four things the fab would like to know. Park has a queue-time rule to write. Every shift will have to follow it without him standing there. Today you settle all three of those. Tomorrow is the last day anything can be measured. After that the recipe is only what is already written down.

BRIEFING 13W

Every step in the deposition sequence has a number against it except one.

WHAT YOU DECIDE 16W

Find the step that governs the whole sequence, and spend what is left of the budget.

13.1 Five steps, one that decides

DEPOSITION BAY • A ROOM • CHAIN

SCENE 31W

Ostrowski has the five steps between the cylinder and the film written up with a measured number beside four of them. The fifth is the one the February controller change moved.

GUIDE 55W

Every step here has to happen for a film to grow, so the sequence is limited by whichever required step is furthest from doing its job. Put them in the order the atoms travel, then name the one that decides the outcome. A step with an alarming-looking number is not automatically the one that governs.

ASK 20W

Put the five steps in the order the atoms take, then name the one deciding whether the film grows correctly.

BEHIND THE BUTTON 134W

Why a sequence is limited by one step. Atoms handed from one step to the next have to get through every step along the way. So the slowest required step sets the rate, however capable the others are. Adding capacity anywhere else changes nothing.

Why the mean free path looks worse than it is. Five centimetres in a thirty-centimetre chamber has been true of this process since it was installed and the film grew correctly for four years. A number that has not changed cannot explain something that has.

What the temperature is for. Breaking the precursor apart takes energy in, and the hot surface is where that energy comes from. Eighteen degrees is a small fraction of the setpoint and a large fraction of the margin above where the reaction runs at all.

TAKES AS READ

every step in a sequence has to carry what the step before it delivers • a reaction that takes energy in needs a source of it

VERDICT 116W

Four of these five steps are doing what they have always done, and the film grew correctly for four years while they did. The mean free path is the number that looks worst, and it is unchanged. That is exactly why it cannot be the explanation: a constant cannot account for a variable. Breaking the precursor apart takes energy in, and the only place that energy comes from is the hot surface. So eighteen degrees below setpoint is a direct reduction in the one step the outcome depends on. A film that grows with too little of it grows with the wrong composition rather than not at all, which is why the tool reported success throughout.

TAKEAWAY 25W

A sequence is governed by whichever required step is furthest from doing its job, and a number that has not changed cannot explain a change.

13.2 Two more pieces of work

WAFER STORE · A ROOM · SCIENCETANK

SCENE 28W

Twill has money for about two more pieces of work before the lot runs and four things people have asked for. Nobody has written down a recommended plan.

GUIDE 47W

Spread a hundred points across the four proposals. Every one of them is defensible; what separates them is which failures they would catch before the lot runs rather than after it. The evidence below carries a fact about each, including the two that are already partly answered.

ASK 12W

Fund the checks that would catch a failure before the lot runs.

TAKES AS READ

a check is worth what it would catch that nothing else would

VERDICT 131W

Two of these four would have caught this failure before the lot ran, and two would only have described it afterwards. The substrate type was never on the incoming certificate, because one supplier shipped one type for thirty years and nobody added a field for it. That is the assumption the fab did not know it was making, and the test takes four minutes on a probe already in the building. The second thermocouple is the other one. The eighteen degrees showed up only in a composition measurement taken six weeks later. So the furnace had no way to say what it was doing at the time. Measuring the failing lots a third time buys nothing — two runs already agree. Photographing the yield maps is a report, not a check.

TAKEAWAY 31W

A check is worth what it catches that nothing else would. A check on a thing that never varied is where a fab finds out it had no field for it.

13.3 The order the steps have to happen in

YELLOW BAY • A PERSON • SEQUENCE

SCENE 40W

Park has to write the queue-time rule down before the lot runs, because he will not be on shift for all of it. Four steps have to happen in an order that keeps the surface bare when the resist lands.

GUIDE 50W

The oxide starts growing the moment the surface is bare and the coater is the step that has to meet it bare. Order the four so that the exposure to air is as short as it can be. What decides the order is which steps can wait and which cannot.

ASK 13W

Order the four steps so the surface is bare when the resist lands.

TAKES AS READ

a bare surface begins to oxidise as soon as it meets air • a step that can be done in advance does not add to any wait

VERDICT 101W

Only one of these four steps starts a clock, and everything after it is inside that clock. Preparing the resist is the one step that can be done at any time. So it goes before the strip rather than after it. Doing it in the wrong order adds its whole duration to the exposure. The rinse and dry has to stay inside the enclosure because leaving it is what the queue time is measuring. A rule written this way survives being followed by somebody who does not know why it is a rule, which is what writing it down is for.

TAKEAWAY 22W

Where a surface changes with time, the order of the steps is part of the process and not a matter of convenience.

END OF THE DAY 21W

A sequence is governed by whichever required step is slowest or weakest, whatever the steps either side of it can do.

Predict it, change it, then measure it

PLAN CARD 108W

Thursday. The qualification recipe is signed at noon. The temperature correction is written into it, and Ostrowski is sure it will work. That is exactly when a fab stops measuring. Aubrey has a copper fill that failed its polish on two wafers overnight, and three panels of data about it. Ganesan has a customer question about why the finished device conducts at all. Today you write down what the corrected recipe should produce, before it runs. Then you find out what actually happened. You also answer for those two failures, which are not what this fortnight has been about. Tomorrow the lot runs on whatever is signed today.

BRIEFING 15W

The recipe change is written and nobody has said in advance what it should do.

WHAT YOU DECIDE 12W

Commit to what the corrected recipe will produce, and then find out.

14.1 What the corrected recipe should produce

DEPOSITION BAY · A ROOM · VERIFY

SCENE 35W

The furnace has been corrected and a wafer is ready to run on the new setpoint. Ostrowski has two hours of analysis time left in the fortnight and has not yet said what he expects.

GUIDE 53W

Lock a predicted composition before the run and it cannot be edited afterwards. Then run it, then spend the analysis time to find out what the film is actually made of. The specified compound fixes what a correct film should be; how close the corrected recipe gets to it is the open question.

ASK 11W

Predict the film composition, make the correction, and report what happened.

BEHIND THE BUTTON 110W

Why the prediction is locked first. With the measurement in hand, an expectation can be adjusted to fit it and the prediction will appear to have been right.

Committing first is what makes the measurement capable of testing anything.

What a confirmed change is not. Raising the setpoint and seeing the controller follow it is a fact about the furnace. What the film is made of is a fact about the reaction, and only a composition measurement establishes it.

Why the analysis time is the constraint. There are two hours left and the lot runs tomorrow, so a measurement not taken today is a measurement about a decision already made.

TAKES AS READ

a film's composition can be measured as a percentage by mass · a stated compound implies a definite composition

VERDICT 84W

The specified compound implies a little over sixty per cent silicon by mass, and that is what the corrected film comes back at. The failure this stop exists to catch is finishing without measuring. A controller that follows a new setpoint is evidence about a controller. The whole fortnight has turned on the difference between a tool reporting success and a layer being the right substance. Predicting first is what makes the measurement a test, because an expectation adjusted afterwards can never be wrong.

TAKEAWAY 22W

A change that has been made and not measured is a change to the tool, not a demonstrated change to the film.

14.2 The fill that did not survive the pad

WET BENCH · A ROOM · DIAGNOSIS

SCENE 34W

Two wafers came off the polish overnight with the copper drawn out of the wide features and intact in the narrow ones. Aubrey has three panels of data and one of them is quiet.

GUIDE 41W

Four explanations are on offer and only one accounts for every panel, including the ones that look ordinary. Read the quiet panels first: a wide feature failing while a narrow one holds is the pattern that rules most of these out.

ASK 9W

Which explanation fits every panel, not just the failure?

TAKES AS READ

a polishing pad presses down on whatever is under it · a metal deforms by planes of atoms sliding over one another

VERDICT 86W

Two panels here are quiet and they do the work. The pad was on recipe, so the polish itself did not change. The narrow features on the same wafer, under the same pad, came out flat. That is what a strength problem in the metal looks like and what a pressure problem does not, since a heavier pad would dish the narrow features too. The metal came in at under a tenth of the specified addition, so the obstructions that interrupt the sliding planes are missing.

TAKEAWAY 27W

An explanation has to account for which features failed as well as that some did, and the pattern across feature sizes is what rules the alternatives out.

14.3 Why the finished device conducts at all

IMPLANT BAY • A PERSON • CASEBOOK

SCENE 32W

The customer's own engineer has sent four statements about the finished device and asked which of them are true. Ganesan wants each one matched to what it actually describes before anybody replies.

GUIDE 47W

Each statement is either true of the material, true of something else, or a confusion between the two. Match them to what each one is really about. Two of them are true sentences about the wrong quantity, which is the commonest way a specification argument goes wrong.

ASK 9W

Match each customer statement to what it actually describes.

TAKES AS READ

a dopant atom brings an electron that was never in a bond • conduction requires charge free to move

VERDICT 86W

All four statements are true, and they are true of four different mechanisms. That is why answering them one at a time is the only honest reply. Doping supplies carriers outright. Heat and light free them out of bonds at a cost, and a metal has them free already. The distinction that matters commercially is the first against the third. The doped silicon conducts because of atoms added on purpose, and it is that control, not the size of the conduction, that makes it a semiconductor.

TAKEAWAY 24W

A statement can be true of the wrong quantity, so matching each claim to what it actually describes is what settles a specification argument.

END OF THE DAY 16W

An intervention is verified by measuring what it changed, not by confirming that it was made.

The lot runs

PLAN CARD 112W

Friday. The qualification lot goes into the yellow bay at ten. Twill has the substrate release form waiting for a signature. There are two lots he will not sign for. The customer's team arrives on Monday. They will read a fortnight of work in twenty minutes, and none of them was here for any of it. Tanabe has one last question about the lamps, sent up by the maintenance contractor. Today you release the substrate for the lot. You decide what goes at the top of the handover. Then you settle what the light in this building can be trusted to do. Nine hundred jobs ride on what goes in at ten.

BRIEFING 14W

The qualification lot goes in this morning and the handover goes out this afternoon.

WHAT YOU DECIDE 18W

Release the substrate, hand over what is known, and say what the light can be trusted to do.

15.1 What goes into the lot

WAFER STORE • A ROOM • CHOICE

SCENE 33W

Twill has fifteen lots in the store, two of them the type nobody had a field for, and a release form for the qualification lot. The probe test takes four minutes a lot.

GUIDE 46W

The four options differ in what they treat as sufficient evidence that a lot is the material the process needs. What separates them is whether the property in question is one the certificate has a field for, and whether magnitude alone can distinguish the two types.

ASK 11W

What is sufficient to release a lot for the qualification run?

TAKES AS READ

resistivity measures how much conduction there is rather than what carries it • a release decision is about the material going in rather than the material already used

VERDICT 87W

The certificate is not wrong; it is silent. Resistivity is a magnitude that both substrate types can carry equally. So a lot inside the band tells you how much conduction there is, and nothing about what is carrying it. The process depends entirely on which. Four minutes on a probe the fab already owns closes that gap for every lot. Past yield is evidence about wafers that have already been through eleven tools, and a film measurement comes weeks after the release decision it would have informed.

TAKEAWAY 22W

A release decision rests on the property the process depends on, which is not always a property anybody wrote a field for.

15.2 What goes at the top

METROLOGY BAY · A PERSON · TRIAGE

SCENE 28W

The customer's team arrives on Monday with twenty minutes to read a fortnight. Ferreira has the spectra, the compositions, the two causes and one finding nobody has chased.

GUIDE 46W

The options differ in what they put in front of somebody who has to act before they can read. What separates them is whether the reader could make Monday's decision from it alone, and whether it says what was not established as well as what was.

ASK 8W

What belongs at the top of the handover?

TAKES AS READ

a receiving team acts before it reads the detail

VERDICT 98W

The reader has to decide on Monday whether to accept the lot. The only thing that answers that is what was found, how firmly, and what is still unexplained. Naming the open carbon finding at the top is not a weakness in the handover. It is the difference between a report and a record. The next person to see that peak needs to know it was seen before. The spectra and the tables are the evidence behind the summary, so they belong underneath it. Somebody can then stop reading as soon as they have what they came for.

TAKEAWAY 21W

A handover is ordered by what the next decision needs, and the unfinished findings belong in it rather than after it.

15.3 What the light can be trusted to do

YELLOW BAY • A ROOM • PROTOCOL

SCENE 32W

The maintenance contractor has four lamps to quote for and Tanabe will not sign for any of them without the reason written down. The lot is in the bay while they talk.

GUIDE 42W

Each lamp differs in wavelength or in power, and only one of those two decides whether the resist responds at all. Match each lamp to what it can be trusted to do. The distinction is the one this bay is lit by.

ASK 11W

Match each lamp to what it can be trusted to do.

TAKES AS READ

the energy of one photon is set by its wavelength • the power of a lamp sets how many photons arrive each second

VERDICT 92W

Two properties of a lamp are being confused with each other everywhere in this conversation. Wavelength fixes what one photon carries, and therefore whether it can do the job at all. Power fixes how many arrive, and therefore how fast a job it can do gets done. That is why a bright long-wavelength lamp is safe at any brightness while a dim short-wavelength one is not safe at all. It is the whole reason a room full of people can work under amber light, beside a tool that would expose in milliseconds.

TAKEAWAY 26W

Wavelength decides whether light can do a job at all and power decides how fast it is done, so the two cannot substitute for each other.

END OF THE DAY 16W

A handover states what is known, how well it is known, and what was never established.

THE CLOSING CARD

The qualification lot went in at ten on the Friday and came out at ninety-one per cent, which is two points above the number the customer had written into the contract and four above what Fab 7 was managing before March. Two substrate lots went back to the supplier, who added a type field to the certificate the following quarter for every customer they have. The furnace controller was replaced again, and a second thermocouple now reports beside it on a separate logger.

What it cost: eleven weeks of wafers written off before anybody measured the right thing, a supplier relationship that will be cooler for a year, and four days of spectrometer time that went on wafers already scrapped. What is unfinished: the carbon layer under the nitride has a source nobody has found, the queue-time rule is a piece of paper rather than an interlock, and the copper is still being bought against a specification that has been met at a tenth of its stated value twice.

And the reason there was a lot to run at all is that you would not let one explanation stand for two faults. You opened the four measurements that agreed and found what they had all been compared against; you took the furnace seriously after the substrate had already been proved wrong; and you predicted what the corrected recipe would produce before you ran it, which is the only reason anybody can say it worked. Nine hundred people came back on the Monday. That was your fortnight.

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